Instruction set specification

Five-Stage Pipelined Microcontroller for a Washing Machine
Logisim-Evolution 5.0.0 • Assignment 2

1 Exactly three instruction formats

Format	Bit allocation	Operations / example
R	15:12 op | 11:9 Rd | 8:6 Rs | 5:0 = 0	MOV, ADD, SUB, AND, LD, ST, CMPGEU; 3280 = ADD R1,R2
I	15:12 op | 11:9 R | 8 = 0 | 7:0 immediate	LDI, ADDI, BEQZ, BNEZ; 1464 = LDI R2,100
J	15:12 op | 11:8 = 0 | 7:0 target	JMP; B020 = JMP 0x20. NOP/HALT require all low 12 bits = 0.

2 Opcode and decode table

Op	Mnemonic	Reads in EX	Writes / operation
0	NOP	none	No architectural write
1	LDI R,imm	none	R = imm
2	MOV Rd,Rs	Rs	Rd = Rs
3	ADD Rd,Rs	Rd, Rs	Rd = (Rd + Rs) mod 256
4	SUB Rd,Rs	Rd, Rs	Rd = (Rd - Rs) mod 256
5	AND Rd,Rs	Rd, Rs	Rd = Rd AND Rs
6	LD Rd,[Rs]	Rs	Rd = DATA[Rs]; result at end of MEM
7	ST Rd,[Rs]	Rd, Rs	DATA[Rs] = Rd; single write in MEM
8	ADDI R,imm	R	R = (R + imm) mod 256
9	BEQZ R,target	R	If R = 0, PC = target
A	BNEZ R,target	R	If R != 0, PC = target
B	JMP target	none	PC = target
C	HALT	none	Stop fetch; drain older entries; reset to resume
D	CMPGEU Rd,Rs	Rd, Rs	Rd = 1 if unsigned Rd >= Rs; otherwise 0
E/F	Unused	none	Deterministic NOP

All listed reserved bits must be zero. Any malformed encoding becomes NOP, including malformed branches and HALT: no register, RAM or I/O write, no redirect, and no halt. All eight registers, including R0, are writable.

3 Execution semantics and the custom instruction

PC is an 8-bit instruction-word address. Reset selects word 00. Sequential execution adds one modulo 256; branches and JMP use an absolute 8-bit target, without a delay slot. Instructions are 16 bits, so the PC is not shifted before addressing ROM. LDI copies eight bits exactly. ADDI 0xFF subtracts one modulo 256; there is no immediate extension or overflow trap.

CMPGEU supports robust water-level tests even when sensor samples skip the exact threshold. Fill tests WATER_LEVEL >= target. Drain loads 10 into a temporary register, then tests 10 >= WATER_LEVEL. A sign-bit test after subtraction would be incorrect for general unsigned bytes.

The gate-built ripple adder performs A + NOT B + 1. Its final carry means no unsigned borrow, hence A >= B. CMPGEU returns this carry in bit 0 and zeros in bits 7:1. Example D280 is CMPGEU R1,R2. It replaces a multi-instruction unsigned comparison routine with one ALU instruction and one branch.

Both source values are consumed in EX. Result availability is the same as ADD: at the end of EX. EX/MEM and MEM/WB forwarding apply normally; a preceding load needs one bubble. No flags register or extra pipeline stage is added.

4 Control signals

Signal	**Instructions asserting it**
RW	LDI, MOV, ADD, SUB, AND, LD, ADDI, CMPGEU
LD / ST	LD / ST respectively
UD	ADD, SUB, AND, ST, ADDI, BEQZ, BNEZ, CMPGEU
US	MOV, ADD, SUB, AND, LD, ST, CMPGEU
BZ / BN / JP / HT	BEQZ / BNEZ / JMP / HALT respectively

UD means the Rd/R field is a real source. US means the Rs field is a real source. A destination-only LDI must not stall after LD merely because its destination matches; the provided source_use test verifies this.

5 Machine-code examples

Instruction	**Hex**
LDI R0,0xF4	10F4
ST R1,[R4]	7300
BEQZ R1,0x20	9220
ADDI R1,0xFF	82FF
HALT	C000
Malformed HALT: treated as NOP	C001